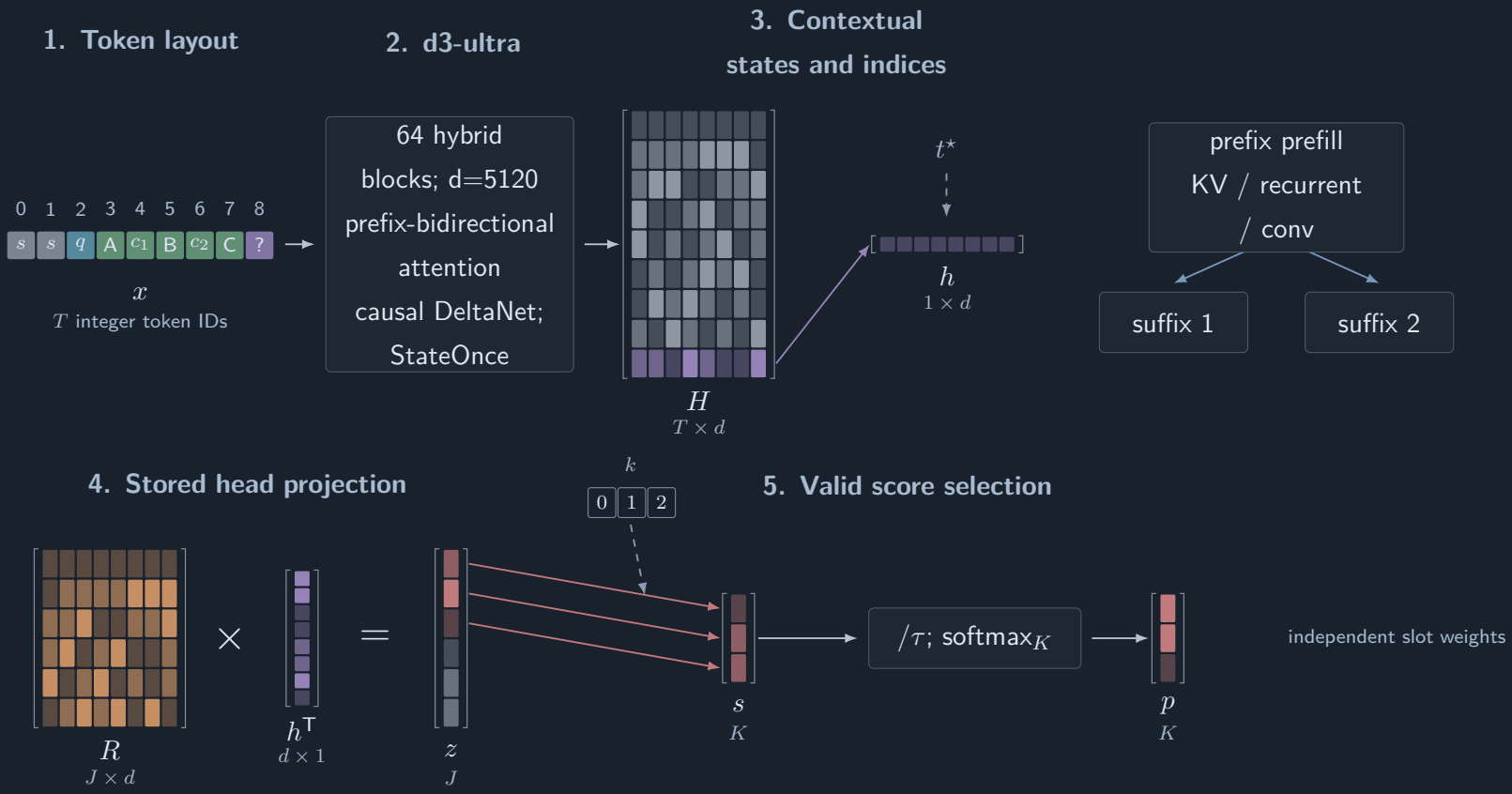
$$h = H_{t^\star,:}, \quad z = Rh^\mathsf{T}, \quad s = z_{0:K}, \quad p = \mathrm{softmax}_K(s/\tau)$$



Axes	$d = 5120$, $J = 255$; Single-question illustration; batch omitted. Tile counts encode axes, not literal sizes.
Objects	Gray: input/context; violet: hidden states; orange: parameter matrices; red: logits/probabilities. Positions, code IDs and owner maps are integers, not scores. Tiles show roles, not measured magnitudes.
Mechanism	Prefix prefill saves KV, recurrent and convolution state; each suffix forks that state. A suffix sees the prefix and its own text, never another suffix. Independent 255 x 5120 readout; initial head and training objective undisclosed. The suffix-end hidden state is read after final normalization.